Manan Tuteja

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Summary

Graduate Mechanical Engineering student specializing in robotic design and control systems. Experience bridging mechanical and electrical CAD, with basic practical programming skills.

Experience

Novartis Pharmaceuticals R&D — Jun. 2026 – Dec. 2026

Robotics Mechanical Engineering Intern

- Development of a fully autonomous **6-DOF robot** for **pick and place** samples using a Beckhoff PLC with **TwinCAT** on **EtherCAT** protocol. Integrated custom **HMI** interface for use in a laboratory environment.
- Implemented state machine on **PLC** through state transition table and state machine diagram.
- Custom end effector design through **Creo** using **resin 3-D printer** with custom mounting for components made with **Onyx Markforged** filament. Created engineering drawings for custom milled parts using **GD&T** and **ASME** standards.
- Developed pneumatic and electrical circuitry integrating vacuum systems, solenoid valves, regulators, gauges, peristaltic pumps with power supplies, circuit breakers, switches, and relays through Visio.

Triton Robotics @ University of California San Diego — Jan. 2024 – Present

Mechanical Team Lead — *Jul. 2025 – Present*

- Robot developed for ARC Robotics Competition with computer vision based turret system.
- Managed a team of over 10 mechanical engineers and general management of 30 member mechanical team in **Onshape** enterprise.
- Directed development of shooting mechanism and pitch mechanism made with flywheels, spur gears, and sensors such as encoders and IMUs.
- Designed and manufactured indexer, ball path, customizable yaw mechanism, and a rectangular x-drive mecanum chassis with tangential suspension.
- Developed custom **4-layer PCB** breakout board for STM32 Nucleo-F446RE with **EasyEDA STD** including SPI, CAN Transceiver, UART, I2C, PWM, and Analog connections.

Mechanical Team Member — *Jan. 2024 – Jun. 2025*

- Robot developed for Robomasters North America Competition with computer vision based turret system.
- Designed **linear suspension system** emphasizing cost-effectiveness and assembly efficiency with **Solidworks**.
- Designed **octagonal omniwheel x-drive chassis** with custom polycarbonate electronics plate.

University of California San Diego CENG — Sep. 2025 – Dec. 2025

CENG 15R Reader

- Reader/Grader for Engineering Computation Using MATLAB for 150+ Chemical and Nano Engineering Students.

Education

University of California San Diego — La Jolla, CA

M.S. Mechanical Engineering — *Sep. 2026 – Dec. 2027*

University of California San Diego — La Jolla, CA

B.S. Aerospace Engineering — *Sep. 2023 – Jun. 2026*

- Cumulative GPA: 3.818
- Major GPA: 3.750

Technical Skills

CAD: Creo, Onshape, Autodesk Inventor, Autodesk Fusion 360, Solidworks, GD&T, ASME Y14.5
Programming: Matlab/Simulink, Python (pandas, NumPy, Matplotlib, OpenCV and ROS2), C++ (Arduino), RaspberryPI, PLC (TwinCAT)
Manufacturing: Lasercutting, Waterjetting, Manual Mill, General Machine Shop tools
PCB Design: EasyEDA